பயிற்சி 10.4

(1) $x^{\cos x}\left(-\sin x \log x + \frac{\cos x}{x}\right)$ (2) $x^{\log x}\left(\frac{2\log x}{x}\right) + (\log x)^x\left[\frac{1}{\log x} + \log(\log x)\right]$

(3) $\frac{y(2x-1)}{x(1+2y)}$ (4) $\frac{y(x\log y - y)}{x(y\log x - x)}$ (5) $(\cos x)^{\log x}\left[\frac{\log(\cos x)}{x} - \tan x \log x\right]$

(6) $-\frac{b^2 x}{a^2 y}$ (7) $\frac{x\sqrt{x^2+y^2}+y}{x - y\sqrt{x^2+y^2}}$ (8) $\frac{1-\sec^2(x+y)-\sec^2(x-y)}{\sec^2(x+y)-\sec^2(x-y)}$

(10) $\frac{1}{2}$ (11) $\frac{6}{1+9x^2}$ (12) 1 (13) $-\tan t$

(14) $\tan t$ (15) $\frac{t^2-1}{2t}$ (16) $\frac{2}{1+x^2}$ (17) $\frac{3}{\sqrt{1-x^2}}$

(18) 1 (19) $\cos x^2$ (20) 2 (21) $\frac{1}{2}$

(22) –1 (23) $\frac{x}{(1-x^2)^{\frac{3}{2}}}$

பயிற்சி 10.5

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
2	4	3	3	1	4	3	2	1	4	3	3	2	2	4
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
2	3	1	4	2	4	1	1	3	2					

பயிற்சி 11.1

(1) (i) $\frac{x^{12}}{12}+c$ (ii) $-\frac{1}{6x^6}+c$ (iii) $\frac{3}{7}x^{\frac{7}{3}}+c$ (iv) $\frac{8}{13}x^{\frac{13}{8}}+c$

(2) (i) $-\cot x + c$ (ii) $\sec x + c$ (iii) $-\text{cosec}x + c$ (iv) $\tan x + c$

(3) (i) $12^3 x + c$ (ii) $\log|x| + c$ (iii) $e^x + c$

(4) (i) $\tan^{-1} x + c$ (ii) $\sin^{-1} x + c$

பயிற்சி 11.2

(1) (i) $\frac{(x+5)^7}{7}+c$ (ii) $\frac{1}{9(2-3x)^3}+c$ (iii) $\frac{2}{9}(3x+2)^{\frac{3}{2}}+c$

(2) (i) $\frac{-\cos 3x}{3}+c$ (ii) $-\frac{\sin(5-11x)+c}{11}$ (iii) $-\frac{\cot(5x-7)}{5}+c$

(3) (i) $\frac{1}{3}e^{3x-6}+c$ (ii) $-\frac{e^{8-7x}}{7}+c$ (iii) $-\frac{1}{4}\log|6-4x|+c$

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